

# Martin Meinel

POSTDOCTORAL RESEARCHER · MACHINE LEARNING FOR HEALTH

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## Profile

Postdoctoral researcher with expertise in leveraging genetic and multi-omic data to inform diagnostics and early-stage target discovery. Developed a diagnostic classifier distinguishing early mycosis fungoides from eczema and psoriasis using bulk, single-cell, and spatial transcriptomic data, now translated into the **patent-pending Erysk-TargetDisk RUO assay**, available through Dermagnostix GmbH. Complementarily, developed methods to identify candidate causal genes and the cellular context in which their regulation is strongest by integrating genetic and single-cell data, advancing precision therapeutic targeting. This work was selected for oral presentation at the NeurIPS 2025 AI4D3 workshop and was supported by the Chan Zuckerberg Initiative.

## Research Experience

### Developed gene-expression based diagnostic classifier for early mycosis fungoides

Helmholtz Munich

PHD STUDENT

Oct 2021 - Aug 2026

- Identified 6-gene biomarker panel discriminating early mycosis fungoides from eczema and psoriasis from bulk RNA-seq data using SAFARI, a computational framework developed for robust biomarker discovery.
- Reduced the biomarker panel to a 2-gene classifier with the option to defer clinically ambiguous cases, achieving **91% sensitivity and 94% specificity** in the qPCR development cohort.
- Validated classifier generalizability across **8 independent international medical centers**.
- Showed earlier detection than standard-of-care methods (histopathology and clinical presentation) in four longitudinally followed patients.
- Characterized marker expression in single-cell and spatial transcriptomics (10x Visium) data by integrating public datasets, revealing *LCK* expression in T cells (MF) and *HOMER1* in keratinocytes (eczema, psoriasis).

### Causal inference for early-stage target discovery (FocusMR)

Helmholtz Munich, CZI Residency

PHD STUDENT

Oct 2023 - Aug 2026

- Identified candidate causal disease genes with the cell states where regulation is strongest using FocusMR, an attention-based single-cell Mendelian Randomization framework.
- Prioritized 34 candidate causal eczema genes at cell-type level resolution with **four loci mapping to fine-grained sub-cell-states** by integrating eczema GWAS and single-cell genetics data (OneK1K).
- Demonstrated biological plausibility of the discovered cell states by recovering known disease-associated biology, including *PRKCQ-AS1* in naive/central-memory CD8<sup>+</sup> T cells enriched for PD-1 signaling and CD28 costimulation.
- Resolved a *TAPBPL*-defined CD4<sup>+</sup> T cell state that **spans several annotated subtypes** rather than localizing to one, enriched for a shared Th17/Th22 program.
- Developed a Bayesian strategy to assess robustness of candidate cell states across independent individuals, supporting context-aware target prioritization.

## Publications & Manuscripts

- Meinel, M.<sup>†</sup>, Langreder, N.<sup>†</sup>, et al. (2026). LCK and HOMER1 gene-expression based classifier distinguishes early mycosis fungoides from eczema and psoriasis. Manuscript in 3rd-round revision at *Science Translational Medicine*. preprint: <https://www.medrxiv.org/content/10.1101/2025.10.18.2533802>.
- Garger, D.<sup>†</sup>, Meinel, M.<sup>†</sup>, Dietl, T., Hillig, C., Garzorz-Stark, N., Eyerich, K., de Angelis, M., Eyerich, S., Menden M.P. (2023). The impact of the cardiovascular component and somatic mutations on ageing. *Aging Cell*, <https://doi.org/10.1111/accel.13957>.
- Fischer, F., Doll, A., Uereyener, D., Roenneberg, S., Hillig, C., Weber, L., Hackert, V., Meinel, M., Farnoud, A., Seiringer, P., et al. (2023). Gene expression based molecular test as diagnostic aid for the differential diagnosis of psoriasis and eczema in formalin fixed and paraffin embedded tissue, microbiopsies and tape strips. *The Journal of Investigative Dermatology*, S0022-202X(23)00156-2. <https://doi.org/10.1016/j.jid.2023.02.015>.
- Farnoud, A., Ohnmacht, A. J., Meinel, M., & Menden, M. P. (2022). Can artificial intelligence accelerate preclinical drug discovery and precision medicine?. *Expert opinion on drug discovery*, 17(7), 661–665. <https://doi.org/10.1080/17460441.2022.2090540>

<sup>†</sup> symbolizes shared first authorship.

## Skills

### Genomic Technologies

Single-Cell RNA-seq analysis & integration (scanpy, Harmony, scVI, scANVI, scCODA)  
Spatial transcriptomics analysis & integration (scanpy, scANVI, 10x Visium)  
Statistical Genetics (Mendelian Randomization, GWAS, HEIDI, Colocalization, PLINK)  
Bulk RNA-sequencing analysis (DESeq2, edgeR, fgsea, clusterProfiler)

### Programming & Data Analysis

Python (NumPy, Pandas, scikit-learn, PyTorch), R, Bash

### High-Performance & Cloud Computing

hydra, Slurm, Docker, Git, AWS

### Languages

English (Fluent), German (Native)

## Education

### Postdoc with focus on context-dependent eQTL mapping

HELMHOLTZ CENTER MUNICH | CASALE LAB

Munich, Germany

Aug 2026 - present

### PhD Candidate in computational biology and statistical genetics

HELMHOLTZ CENTER MUNICH | LMU MUNICH | CHAN ZUCKERBERG INITIATIVE RESIDENCY

Munich, Germany

Oct 2021 - Aug 2026

- Supervised by Dr. Michael Menden, Dr. Natalie Garzorz-Stark & Dr. Francesco Paolo Casale

### M. Sc. in Data Engineering & Analytics

TECHNICAL UNIVERSITY OF MUNICH

Munich, Germany

Oct 2018 - Nov 2020

### B. Sc. in Applied Computer Science

UNIVERSITY OF BAYREUTH

Bayreuth, Germany

Oct 2014 - Oct 2017

## Supervision, Teaching & Review

- 2025 Supervisor of three students for their master's theses and internships.
- 2024 Teaching assistant for Computational Modeling for System Genetics.
- 2025 Reviewer for the **RECOMB** Conference.
- 2024 Reviewer for the **ML for Health** workshop.

Munich, Germany

TU Munich

Seoul, KOR

Vancouver, CA

## Honors & Awards

### Patent

- 2025 Published patent application WO/2025/051923, "Differential Diagnosis of Mycosis Fungoides: Method and System", leveraging SAFARI to discover biomarkers distinguishing early MF from eczema & psoriasis.

Munich, Germany

### Oral presentations

- 2025 Oral presentation at **NeurIPS AI4D3 workshop** on how to identify cellular contexts of causal genes enhancing early-stage drug target discovery.
- 2025 Oral presentation at **EORTC CLTG** about the identification and clinical translation of biomarkers in early MF.

San Diego, USA

Athens, Greece

### Chan Zuckerberg Initiative Residency

- 2024 Selected for CZI residency for developing cell-context-aware target discovery methods to identify cellular contexts of causal genes for precise therapeutic targeting.

Redwood, USA

### Invited speaker session

- 2023 Presented "Biocomputational Analysis of Big Data" to researchers in dermatology at the EADV-ESDR Summer Research Workshop: Big Data and Translation in Dermatology.

Freiburg, Germany

### Poster presentations

- 2025 Highlight poster with oral presentation on cell-state-aware single-cell Mendelian Randomization at **RECOMB Genetics**.
- 2023 Presented the SAFARI framework for rare disease biomarker discovery at the **ML for Health** workshop.
- 2025 Presented work on cell-state-aware single-cell Mendelian Randomization at the **Helmholtz AI conference**.
- 2025 Presented classifier development and validation results for early MF diagnosis from psoriasis and eczema at **DKTK meeting**.

Seoul, KOR

New Orleans, USA

Karlsruhe, Germany

Freiburg, Germany

## Working experience

### Opitz Consulting GmbH

SOFTWARE ENGINEER WORKING STUDENT

Munich, Germany

Oct 2019 - Oct 2020

- Built a Docker skeleton for Apache Zeppelin, enabling users to start a docker container running Apache Zeppelin on OpenShift with one click.
- Created a web application for Allianz IDS GmbH, allowing employees to track, assign, and assess investment orders within the department.

### REHAU Industries SE & Co. KG

SOFTWARE ENGINEER WORKING STUDENT

Bayreuth, Germany

Nov 2017 - Mar 2019

- Developed an ALEXA skill for the 2018 FRONTALE window exhibition, demonstrating several features of the smart window.
- Implemented a web app for Medi Bayreuth to track player performance and fatigue for injury prevention.